



Multi-Agent Reinforcement Learning for Ride-Hailing Cab Vehicle Routing on Urban Road Networks

M.Tech. Thesis Defence

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Problem. Route a small fleet on a discrete urban road graph. Each minute, requests arrive at random pickup nodes; free cabs choose *wait* / *move* / *service*; busy cabs follow shortest paths to drop-off.

Setting. SF Financial District subgraph from OpenStreetMap (Boeing, 2017) (≈ 109 nodes); demand calibrated on the Cabspotting trace (Piorkowski et al., 2009); 60-minute horizon. Train at $N=3$; evaluate at $N \in \{3, 5, 8, 10\}$ (100 paired episodes per N).

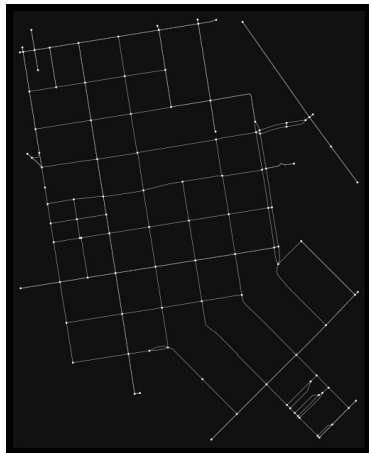
Why MARL. Joint action space $\sim |V|^N$ is intractable; demand is bursty stochastic; each cab sees only a local view. Closely tracks the DP formulation of Garces et al. (Garces et al., 2023).

POSG. $\mathcal{G} = \langle S, \{A_i\}, T, R, \{O_i\}, \gamma, H \rangle$ with $\gamma=0.99$, $H=60$. Action is a masked discrete index of size $2 + \max \deg(G)$.

S : state space; A_i : action set of agent i ; T : transition kernel; R : team reward; O_i : local observation of agent i ; γ : discount; H : horizon.

Team-shaped reward.

$$R = \beta |F| - \lambda |Q^{\text{pre}}|, \quad \beta=15, \lambda=0.05.$$



SF Financial District drive subgraph from OSM.

Unified framework. Every policy exposes (`train_fn`, `load_fn`, `policy_fn`) plus a package tag; orchestration treats all sixteen registered policies uniformly. Paired-seed evaluation aligns metrics across stacks.

Sixteen policies, three pillars.

- **Heuristics (2):** Greedy, Cooperative Greedy.
- **Planners (4):** MCTS-Expert and OneAgentRollout-Expert, each in cost-reward and team-reward variants; thousands of Monte Carlo rollouts per decision under a Coop-Greedy base policy.
- **Learning (10):** Custom **Cent-MAPPO** and **Ind-MAPPO** (Yu et al., 2022; Schulman et al., 2017) (from-scratch on PyTorch+TensorDict, SMDP rollout buffer for partial-decision steps); BenchMARL's (Bettini et al., 2024) MAPPO, IPPO, MASAC, ISAC, QMIX, VDN, IQL; RLlib-PPO (Liang et al., 2018) with the new API stack.

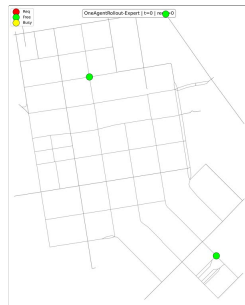
Design choices. CTDE (Lowe et al., 2017) with parameter sharing across agents. Centralised critic sees an agent-agnostic global state (team-presence vector, demand-coverage scalar). Compact hand-engineered local observation ($|V|+2K+\max \deg +9$) and global state ($2|V|+2K+11$). No time-of-episode channel: the policy is trained as a stationary continuing-task agent.

Cross- N deployment. A monkey patch on BenchMARL's MultiAgentMLP lets a policy trained at $N=3$ run unchanged at $N=5, 8, 10$ via parameter-shared per-agent heads.

Cross-algorithm evaluation at $N=3$. 10-episode averages; paired seeds across policies.

Policy	Fulfilled	Wait (Ful.)	Wait (All)
Greedy	15.1	13.97	20.65
Coop-Greedy	15.2	14.08	20.40
MCTS-Expert	13.7	16.24	22.83
OneAgentRollout-Expert	18.8	12.51	20.18
Cent-MAPPO	15.3	13.88	20.63
Ind-MAPPO	15.6	11.51	21.29
BenchMARL-MAPPO	17.2	11.49	20.64
BenchMARL-IPPO	16.9	11.58	20.34
BenchMARL-MASAC	4.7	8.96	26.51
BenchMARL-ISAC	16.0	11.11	21.36
BenchMARL-QMIX	17.0	13.28	20.76
BenchMARL-VDN	18.2	12.09	19.86
BenchMARL-IQL	18.2	12.47	20.16
RLlib-PPO	14.8	13.93	21.36

Bold = column leader; $\sim 29\%$ of ~ 64 demand arrivals fulfilled per episode.



OAR-Expert, seed 42.

Headline findings. The thirteen non-MASAC policies finish within a ≈ 3 -min band on Avg Wait (All) (19.86–22.83 min); adjacent-rank gaps lie within 10-episode noise. BenchMARL-VDN takes Avg Wait (All); OneAgentRollout-Expert takes fulfilment—both only marginally over heuristics. MASAC (4.7 fulfilled, 26.51 wait) is hobbled by cold-start (Papoudakis et al., 2021); MCTS-Expert (13.7, 22.83) shows deep search wastes compute the one-agent rollout commits in one step.

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